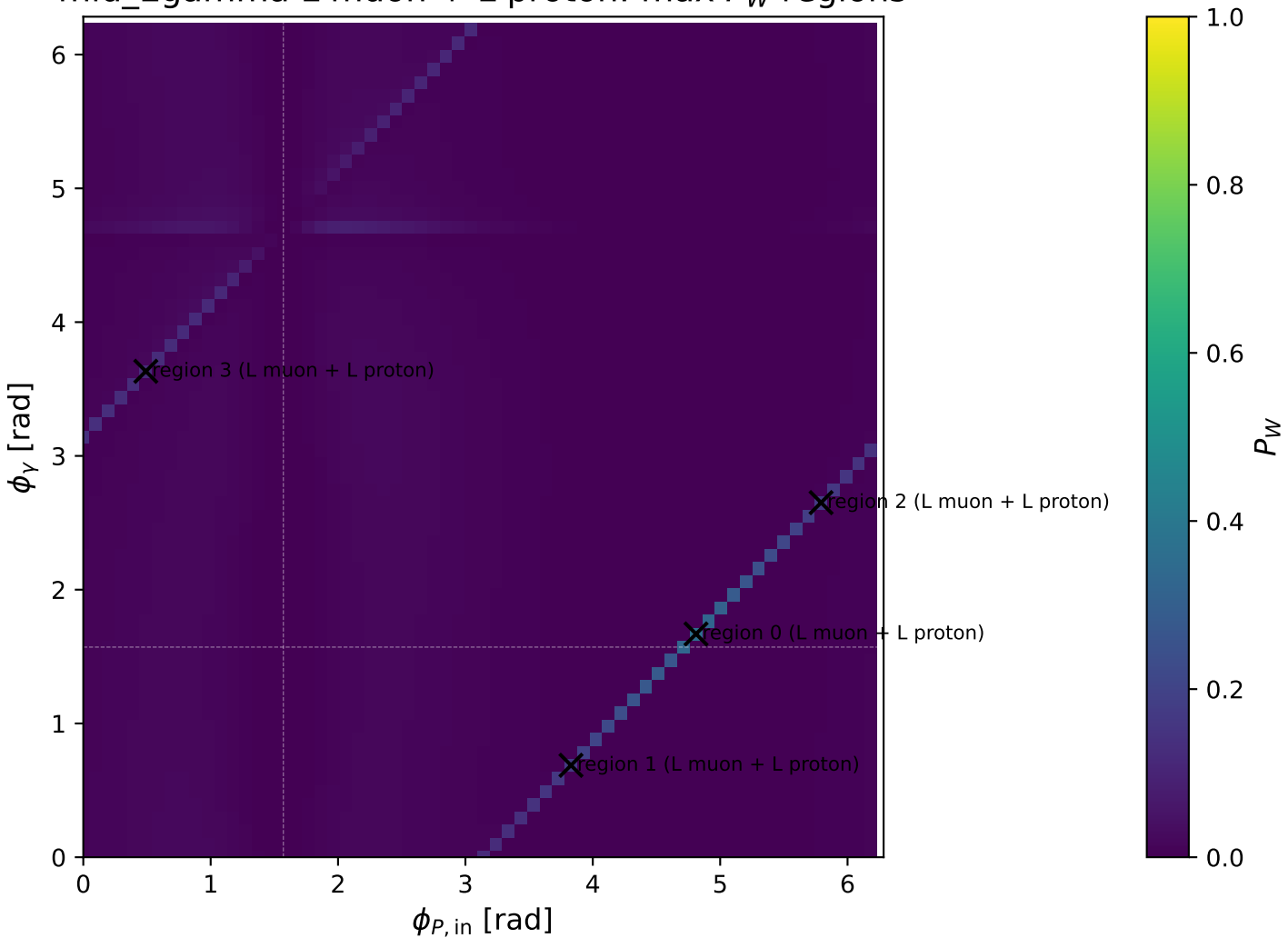
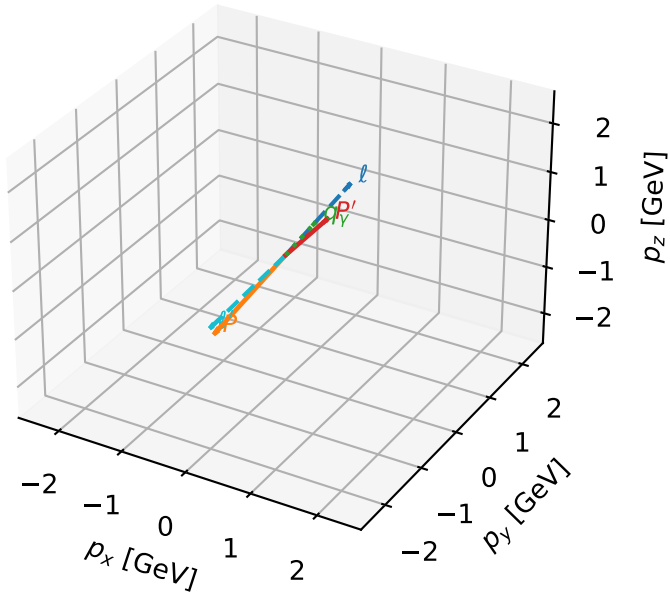


mid_Egamma L muon + L proton: max P_W regions



L muon + L proton characteristic max P_W configuration

3D momenta

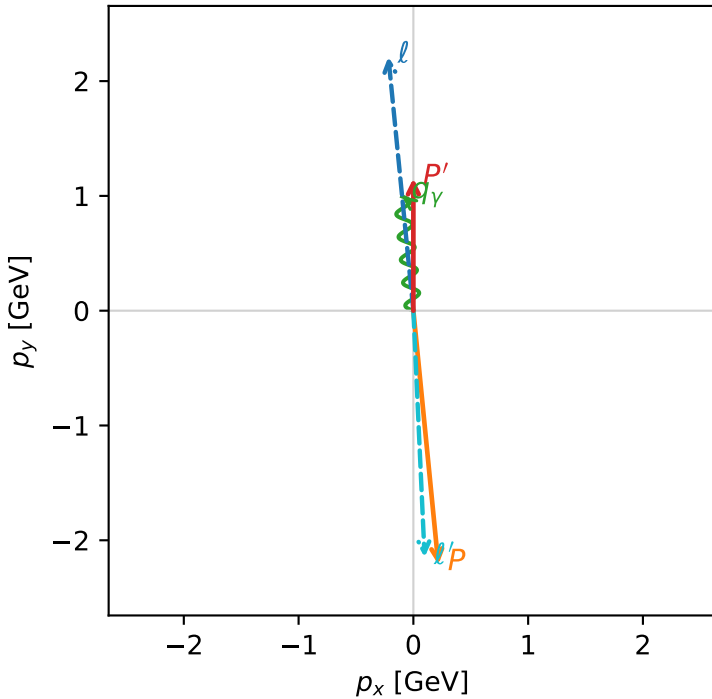


w_purity_mid_egamma_ll_region_0 P_W=0.357701
 region: mid_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

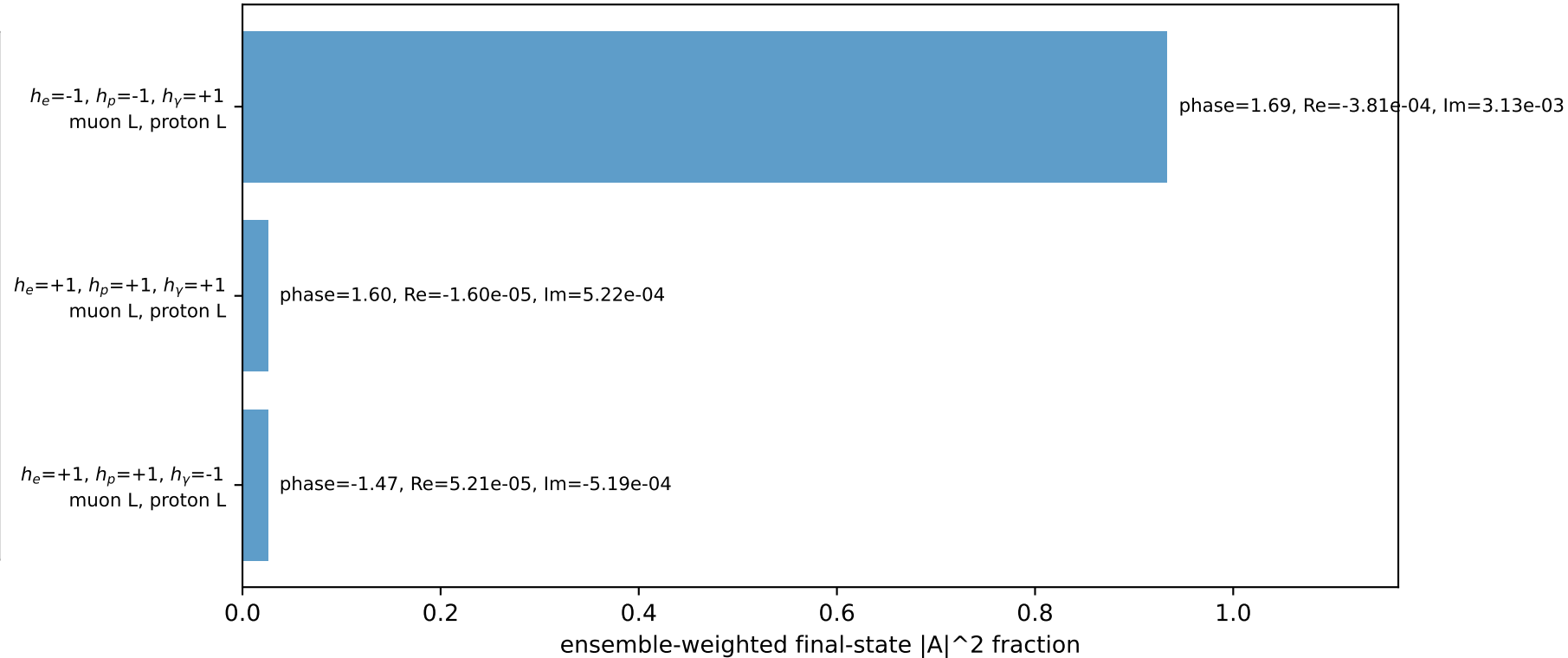
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.15252$
 $\theta_{in}=1.57079$, $\phi_l=1.66897$, $\phi_p=4.81056$
 $E_\gamma=1$, $\phi_\gamma=1.66897$
 $Q^2=19.105$, $x_B=0.965727$, $t=-10.511$, $W^2=1.55787$, $y=0.959186$
 $\theta(l', \gamma)=176.988$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.2179$ $p_y= 2.2124$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.2179$ $p_y= -2.2124$ $p_z= 0.0000$
 l' $E= 2.1525$ $p_x= 0.0980$ $p_y= -2.1477$ $p_z= 0.0000$
 P' $E= 1.4860$ $p_x= 0.0000$ $p_y= 1.1525$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.0980$ $p_y= 0.9952$ $p_z= 0.0000$

Transverse plane

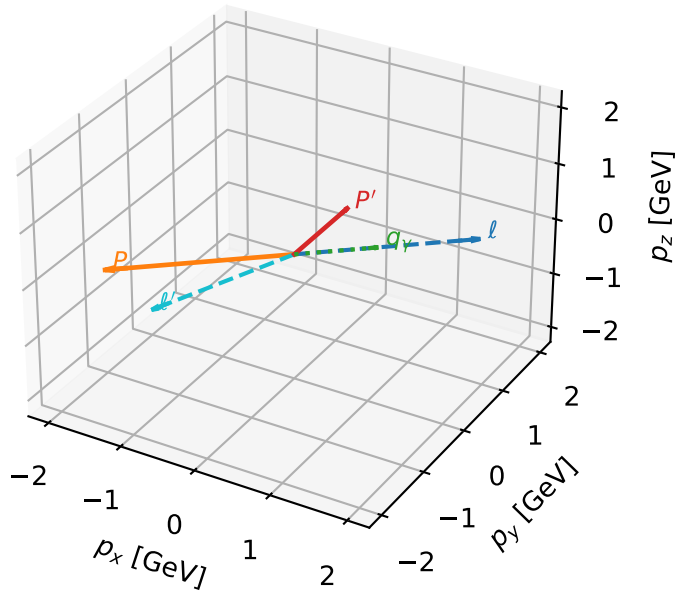


Leading initial-ensemble/final-state components (N=3, retained=98.4%)



L muon + L proton characteristic max P_W configuration

3D momenta

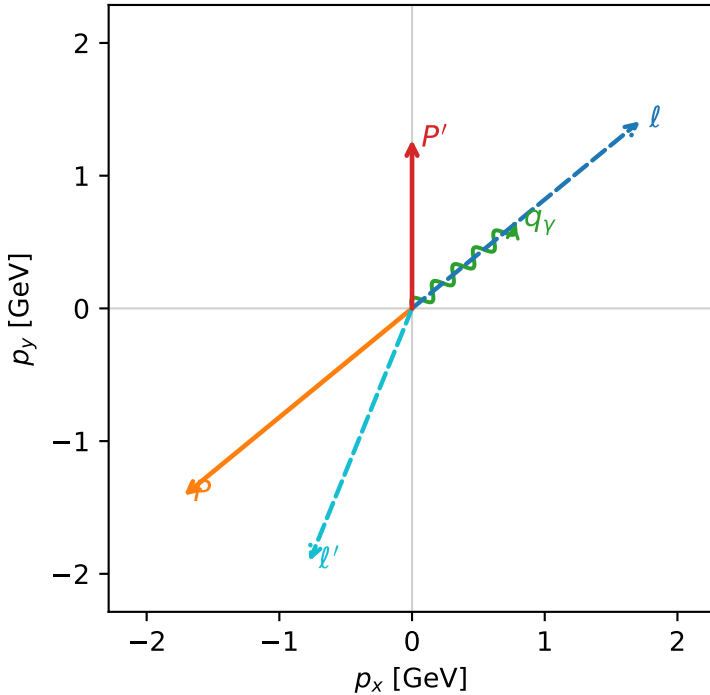


w_purity_mid_egamma_ll_region_1 $P_W=0.174646$
 region: mid_Egamma_LL_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.27096$
 $\theta_{in}=1.57079$, $\phi_l=0.687223$, $\phi_p=3.82882$
 $E_V=1$, $\phi_V=0.687223$
 $Q^2=17.1735$, $x_B=0.91738$, $t=-9.44814$, $W^2=2.4265$, $y=0.907651$
 $\theta(l', \gamma)=151.458$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.7185$ $p_y= 1.4103$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.7185$ $p_y= -1.4103$ $p_z= 0.0000$
 l' $E= 2.0589$ $p_x= -0.7730$ $p_y= -1.9054$ $p_z= 0.0000$
 P' $E= 1.5796$ $p_x= 0.0000$ $p_y= 1.2710$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

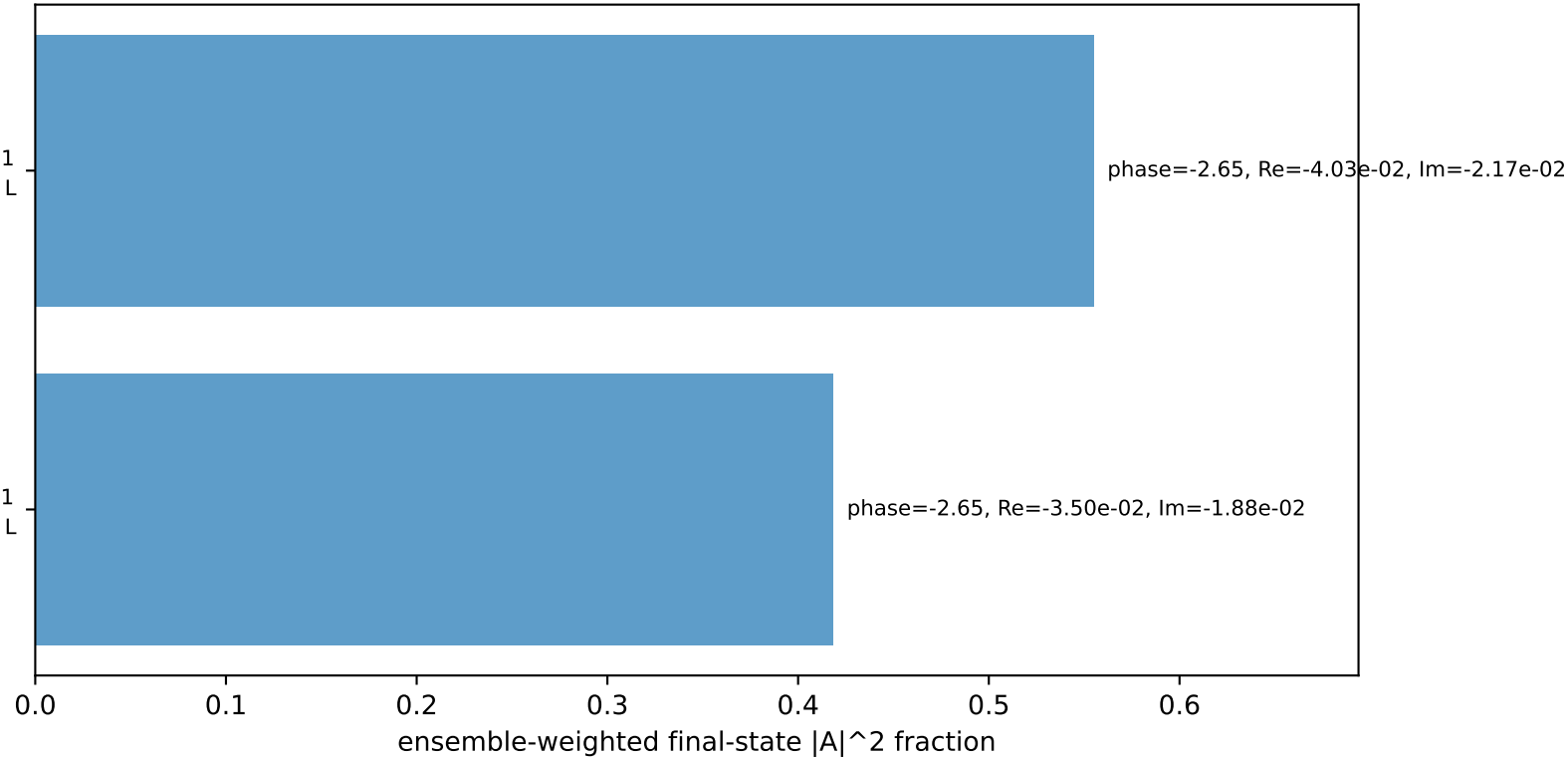
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
 muon L, proton L

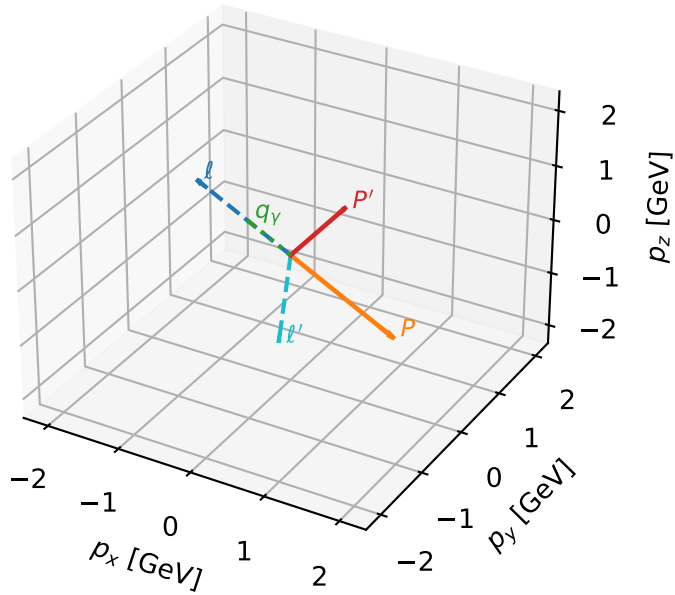
$h_e=-1, h_p=+1, h_\gamma=+1$
 muon L, proton L

Leading initial-ensemble/final-state components (N=2, retained=97.4%)



L muon + L proton characteristic max P_W configuration

3D momenta

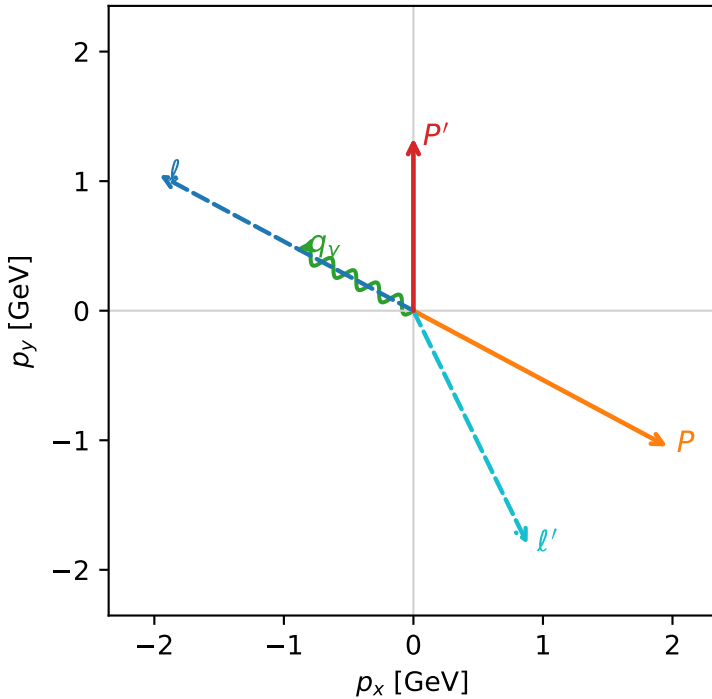


w_purity_mid_egamma_ll_region_2 P_W=0.172248
 region: mid_Egamma_LL_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.33148$
 $\theta_{in}=1.57079$, $\phi_l=2.65072$, $\phi_p=5.79231$
 $E_\gamma=1$, $\phi_\gamma=2.65072$
 $Q^2=16.1607$, $x_B=0.889769$, $t=-8.89083$, $W^2=2.88195$, $y=0.88063$
 $\theta(l', \gamma)=144.192$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.9606$ $p_y= 1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.9606$ $p_y= -1.0480$ $p_z= 0.0000$
 l' $E= 2.0098$ $p_x= 0.8819$ $p_y= -1.8029$ $p_z= 0.0000$
 P' $E= 1.6287$ $p_x= 0.0000$ $p_y= 1.3315$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.8819$ $p_y= 0.4714$ $p_z= 0.0000$

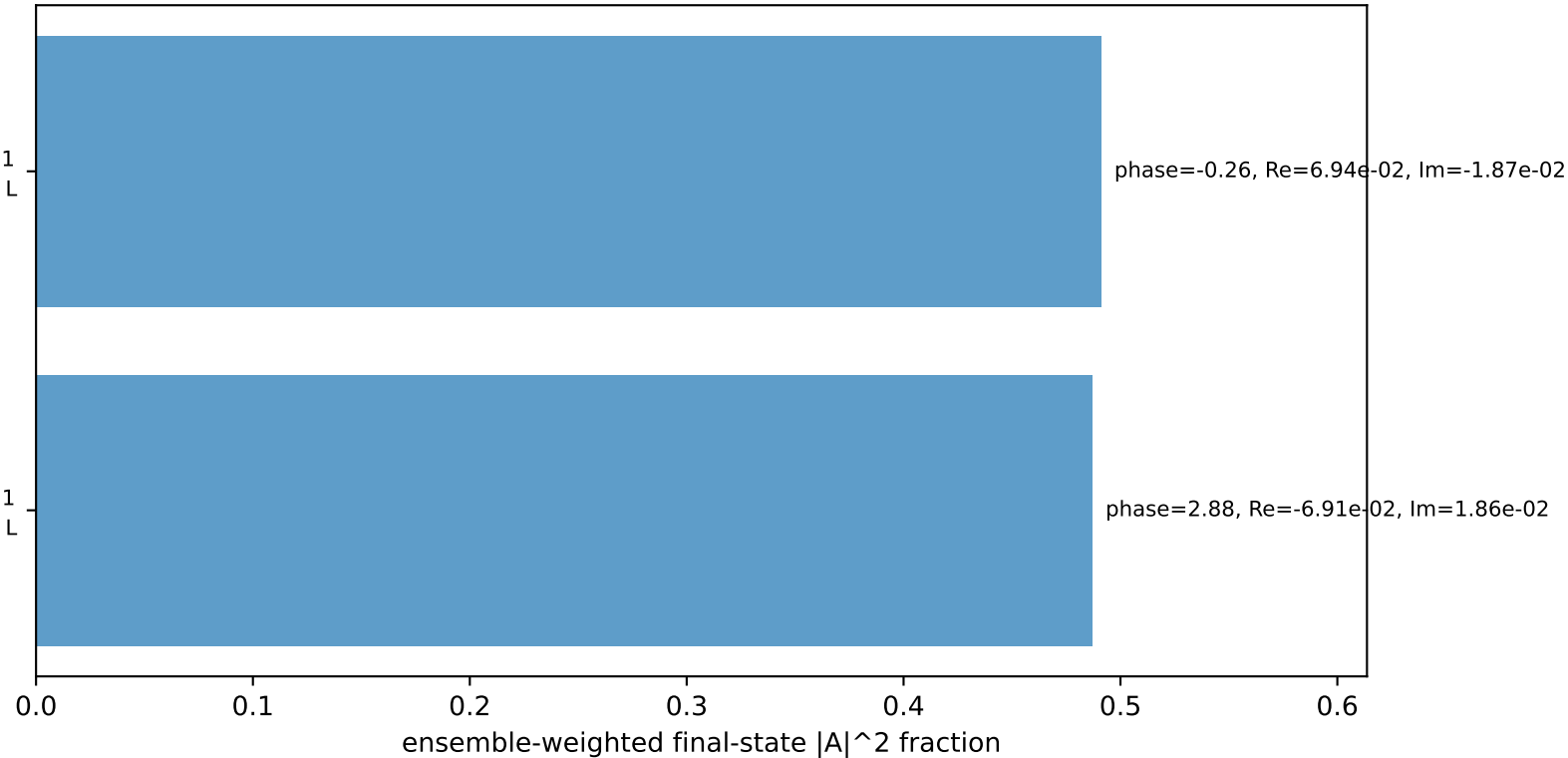
Transverse plane



$h_e=-1, h_p=+1, h_\gamma=+1$
 muon L, proton L

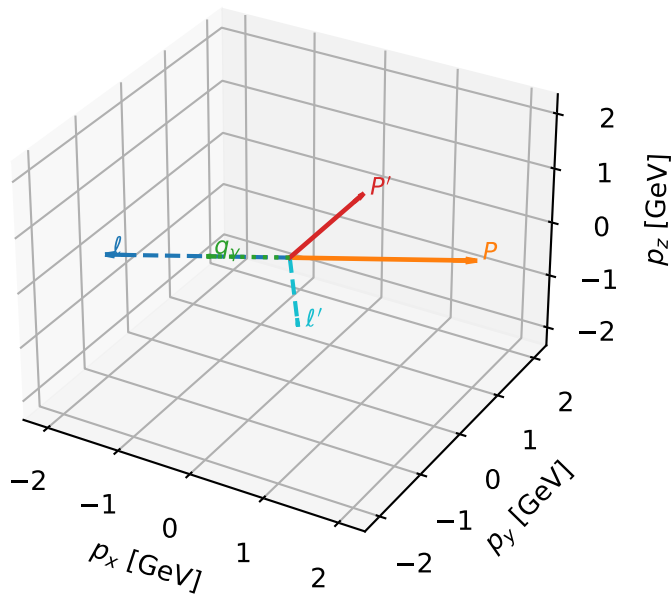
$h_e=-1, h_p=-1, h_\gamma=+1$
 muon L, proton L

Leading initial-ensemble/final-state components (N=2, retained=97.8%)



L muon + L proton characteristic max P_W configuration

3D momenta

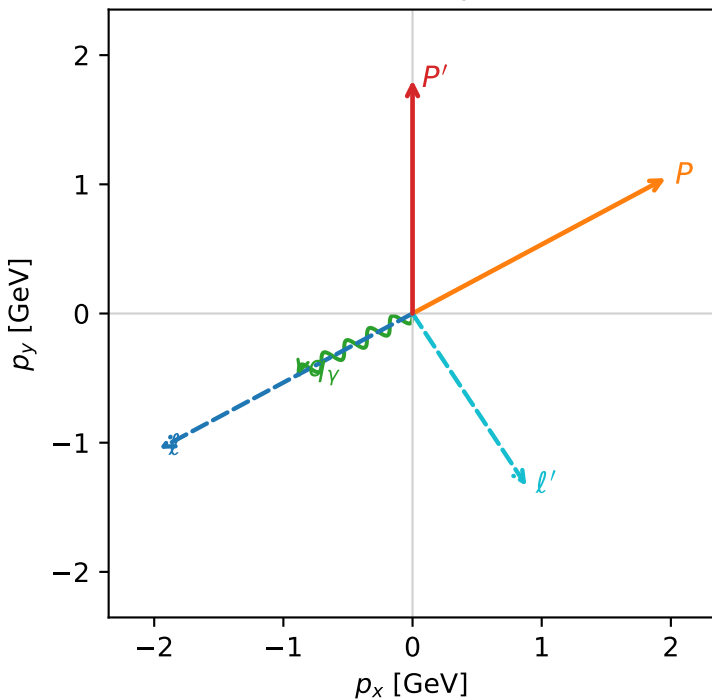


w_purity_mid_egamma_ll_region_3 $P_W=0.124405$
 region: mid_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.80617$
 $\theta_{in}=1.57079$, $\phi_l=3.63247$, $\phi_p=0.490874$
 $E_\gamma=1$, $\phi_\gamma=3.63247$
 $Q^2=7.77496$, $x_B=0.573872$, $t=-4.27622$, $W^2=6.65312$, $y=0.656889$
 $\theta(l', \gamma)=95.3288$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.9606$ $p_y= -1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.9606$ $p_y= 1.0480$ $p_z= 0.0000$
 l' $E= 1.6033$ $p_x= 0.8819$ $p_y= -1.3348$ $p_z= 0.0000$
 P' $E= 2.0352$ $p_x= 0.0000$ $p_y= 1.8062$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.8819$ $p_y= -0.4714$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=98.0%)

